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$f(x, y, z) = xy + yz + zx$ in a $(1, -1, 2)$ in de richting $(1, 1, 1)$

$$\frac{\partial f}{\partial x}(x, y, z) = y + z \Rightarrow \frac{\partial f}{\partial x}(1, -1, 2) = -1 + 2 = 1$$
$$\frac{\partial f}{\partial y}(x, y, z) = x + z \Rightarrow \frac{\partial f}{\partial y}(1, -1, 2) = 1 + 2 = 3$$
$$\frac{\partial f}{\partial z}(x, y, z) = y + x \Rightarrow \frac{\partial f}{\partial z}(1, -1, 2) = -1 + 1 = 0$$
$$\Rightarrow Df(\bar{a}, \bar{h}) = Df(\bar{a})\bar{h} = \begin{pmatrix} 1 & 3 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 4$$

References